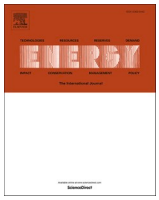
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Decarbonizing rotary kiln–induction furnace based sponge iron production

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ABSTRACT

The direct carbon dioxide (CO_2) emissions from the iron and steel sector are nearly 7 % of the global CO_2 emissions from energy use. India is the world's second-largest producer of steel and the largest sponge iron producer. India produces one-third of the global sponge iron, mostly from rotary kilns using coal as an energy source, resulting in higher energy and emission intensities than the global average. This study evaluates major options for CO_2 abatement in the rotary kiln–induction furnace process for steel production. Based on the mass and energy balances of a typical sponge iron plant, this study evaluates the techno-economic potential of nine major decarbonization measures. The proposed decarbonization measures include improved energy efficiency, improved material efficiency, renewable energy use, and fuel substitution. Marginal abatement and green premium curves are generated by incorporating the dependency of different measures. Depending upon the set of options chosen, carbon abatement is in the range of 55.8 %–99.2 %. It is observed that 11.5 %–20.6 % of decarbonization potential can be achieved without any additional cost to the customer. These findings are insightful for policymakers and industry stakeholders and provide a foundation for designing and implementing a net-zero pathway for the sector.

1. Introduction

Greenhouse gases increase the temperature of the atmosphere, eventually resulting in global warming. Carbon dioxide (CO_2), methane, nitrous oxide, and fluorinated gases are the prominent greenhouse gases. However, CO_2 is the most critical greenhouse gas due to its large scale of emission. Global warming drives climate change, causing more severe and frequent weather conditions, melting ice caps, and other impacts. Reducing greenhouse gas emissions is crucial for mitigating climate change, limiting global warming, improving air quality, and ensuring sustainable use of natural resources.

Energy use contributes to 73.2 % of global greenhouse gas emissions, of which 24.2 % is emitted by the industry sector alone [1]. In 2018, the world manufactured 1.82 Gt of crude steel, making it one of the largest CO_2-emitting sectors [2]. Specifically, direct emissions from the iron and steel sector accounted for 2.6 Gt of CO_2, representing approximately 7 % of global CO_2 emissions generated by energy systems [3]. On an average, for each tonne of crude steel produced in 2018, the industry emitted approximately 1.81 tonnes of CO_2 and consumed around 19.51 GJ of energy [4]. Three primary processes used for crude steel production are the blast furnace–basic oxygen furnace process, the direct reduced iron–electric arc furnace process, and the scrap–electric arc furnace process with specific energy consumption of 24.4, 25.3, and 10.0 GJ per tonne of crude steel (tcs) and specific emission intensity of 2.32, 1.65, 0.67 t CO_2 per tonne of crude steel (tcs) [4]. Many processes, like rotary kiln–induction furnace process, hearth furnaces, fluidized bed furnaces, etc., have a small share in global production and are limited to specific geographical locations. Identifying different CO_2 abatement measures to decarbonize the global iron and steel industry is essential.

Various measures are proposed to decarbonize the iron and steel sector. Kirschen et al. [5] studied the impact of raw materials on the energy balance in the electric arc furnace process for steel making. Dijk et al. [6] discussed sorption-enhanced water gas shift technology for CO_2 capture in the blast furnace process. Tian et al. [7] investigated the role of industrial carbon capture and storage in decarbonizing the steel industry. Arens et al. [8] analyzed several countries based on the size of coal-based steel production and studied the preparedness of countries for a low-carbon transition based on specific indicators, including implementation of low-carbon electricity and climate and energy policy. Fan and Friedmann [9] discussed technology options and their economic implications for decarbonizing these industries for several iron and steel-making processes. Wang et al. [10] discussed the energy efficiency stagnation in global steel production and called for abstinence from primary steel-making to achieve climate targets. Tanzer et al. [11]

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investigated the combined potential of bioenergy and carbon capture and storage technologies to achieve net carbon-negative steel production through various routes. Vogl et al. [12] performed a techno-economic analysis of adopting hydrogen as a reducing agent in steel production and concluded that the total energy demand remains similar to that of the traditional steelmaking route. Nurdiawati et al. [13] performed a life cycle assessment of a bio-syngas-based sponge iron production process combined with carbon capture and storage and compared it with existing processes.

Rootzén and Johnsson [14] explored the limits of CO_2 emission abatement in the European Union's (EU) power and industry sectors, including the steel sector. Li and Zhu [15] analyzed the cost of energy savings and emissions reductions in China's iron and steel industry. Yu et al. [16] analyzed the importance of industrial symbiosis in Chinese integrated steel plants and quantified its emission reduction potential. An et al. [17] studied the possibilities of energy savings and emissions reduction in the Chinese iron and steel industry, considering cost minimization. Chen et al. [18] assessed CO_2 recycling and utilization technologies for low-carbon iron and steel production in China. Wen et al. [19] performed a quantitative analysis of the Chinese iron and steel industry through energy conservation and emission abatement pathways. Griffin and Hammond [20] compared various decarbonization measures for the UK iron and steel industry. Skoczkowski et al. [21] analyzed the decarbonization options in the EU steel industry via a technology innovation method. Toktarova et al. [22] discussed the low-carbon transition pathways for the Swedish steel industry through biomass, carbon capture and storage, energy efficiency, and hydrogen. Gielen and Moriguchi [23] analyzed the emission reduction potential of the Japanese iron and steel industry. There is extensive literature on the decarbonization of the iron and steel industry in China [24] and Europe [25], while there are limited studies for the Indian iron and steel sector, as discussed below.

Elango et al. [26] evaluated net zero scenarios for various processes practiced in the Indian steel industry. Tank et al. [27] proposed the use of the rotary hearth furnace for sponge iron production from solid waste. Prasad et al. [28] proposed process integration techniques in sponge iron plants to utilize heat from kiln off-gas, hot sponge iron, and flue gas. Dey et al. [29] deliberated the opportunities for energy conservation through Pinch Analysis and heat integration of a rotary kiln process for sponge iron production. Dhar et al. [30] created different scenarios for the Indian iron and steel industry and analyzed the transformation in a sustainable 1.5 °C world based on the Kaya identity of the sector. Nduagu et al. [31] compared the lifecycle greenhouse gas emissions of sponge iron production from coal and natural gas-based processes. Bhardwaj et al. [32] investigated decarbonization measures for crude steel production via rotary kiln-induction furnace process. Ghosh et al. [33] presented techno-economic details of various energy-efficient technologies for the decarbonization of Indian rotary kiln-based sponge iron industries. Vigneswaran et al. [34] studied the decarbonization of Indian sponge iron production by replacing coal with biochar. Hall et al. [35] discussed the challenges, transition pathway, and proposed action plan for decarbonizing the steel industry in India.

Some of these papers have discussed a single option of decarbonization, including waste heat recovery [36], the use of biomass products [37], and heat integration. Other research papers considered the iron and steel sector in totality [38] and excluded the coal-based sponge iron sector. Limited literature sources have discussed the combined impact of several decarbonization options on the rotary kiln-induction furnace process and their economics. This paper presents a study about the decarbonization of the rotary kiln-induction furnace process, as it is one of the significant contributors to Indian steel production. The following are the major contributions of this paper.

- Determining emission and energy intensities of a typical rotary kiln-induction furnace based steel plant through mass and energy balances;
- Identification of major CO_2 abatement options, categorized as energy conservation, material replacement, fuel substitution, and utilization of renewable energy;
- Techno-economic analysis of major decarbonization measures is presented based on the mass and energy balance calculations;
- Investigated the expected decarbonization potential of major options, along with their associated costs and the impact of implementing multiple CO_2 abatement options in conjunction with each other;
- Results are presented graphically through marginal abatement and green premium curves; and
- The extent to which decarbonization can be achieved without a net cost to the customer is also analyzed and presented.

The remainder of this paper is organized as follows. The Indian iron and steel sector, with a special emphasis on sponge iron production, is discussed next. The mass and energy balances of a typical rotary kiln-induction furnace based steel plant are presented in section 3. Major decarbonization options are discussed in section 4. Subsequently, marginal abatement and green premium curves are presented in section 5. Finally, the conclusions of the study are presented.

2. Indian iron and steel sector

The Indian iron and steel sector is the second largest in the world, producing 125 Mt of crude steel in 2022, or 6.6 % of the worldwide crude steel output [39]. The Indian iron and steel sector is known for its diversity, with different production processes significantly contributing to the overall output. Specifically, the blast furnace–basic oxygen furnace process contributes about 44.5 % of the total production, the rotary kiln–induction furnace process contributes 19.6 %, and the shaft furnace–electric arc furnace process contributes 5.1 % [39]. The remaining production is achieved through various methods, such as a combination of blast furnaces and electric arc furnaces, electric furnaces for scrap processing, and alternative processes like Corex. The Indian iron and steel sector is characterized by high energy consumption, ranging from 25.1 to 27.2 GJ/tcs [40], and significant CO_2 emissions, with an emission intensity of about 2.3 tonnes of CO_2 per tonne of crude steel [3].

Globally, India is the largest sponge iron producer, accounting for approximately one-third of the global sponge iron production of 111 Mt in 2019 [41]. In contrast to different regions of the world where natural gas is predominantly used in shaft furnaces for sponge iron production, the coal-based process contributes to 80.1 % of India's sponge iron production [39]. Coal is used to produce syngas in a gasifier and then used for the reduction of iron ore lumps/pellets in a shaft furnace. Alternatively, non-coking coal is used in rotary kilns along with iron ore and dolomite as feed materials. These rotary kiln-based sponge iron plants are often accompanied by a power plant to recover heat from the high-temperature kiln-off gases having low calorific value. In the Indian steel sector, sponge iron plants typically use induction furnaces to produce crude steel from sponge iron.

The Indian sponge iron sector is thriving primarily due to the abundant availability of domestic high-grade iron ore, sponge iron grade non-coking coal, low capital investment, ease of sponge iron plant implementation, and better quality of products. Due to the local availability of raw materials, nearly 86 % of India's rotary kiln-based sponge iron capacity is located in five states: Odisha, Chhattishgarh, Jharkhand, Karnataka, and West Bengal. The estimated average capacity of a sponge iron plant is 350 tonnes per day, while the capacities of sponge iron rotary kilns vary between 100 and 500 tonnes per day. Smaller units and plant sizes and the use of low-grade non-coking coal are some of the reasons for the higher energy and emission intensities of the Indian sponge iron industry.

Typically, coal-based sponge iron plants are preferred over gas-based plants in the Indian market due to the lower carbon content in the final

product. Good-grade sponge iron from the rotary kiln process has a metallization of higher than 88 % and a metallic content of more than 80 %. Details of the chemical composition of various grades of coal-based sponge iron are presented in Table 1.

3. Process description with mass and energy balances

The rotary kiln-based sponge iron unit consists of a rotary kiln, a rotary cooler, a magnetic separator, and a system of conveyors for material handling. The rotary kiln is a long cylindrical rotating drum slightly inclined to the horizontal, internally lined with refractories, and has several port holes located along its length for air injection. Feed material containing iron ore (hematite), feed coal, and dolomite is fed into the rotary kiln from the feed end (higher end), and injection coal (also called slinger coal) is injected through the discharge end (lower end). Air is supplied through the lower end and port holes located along the kiln length. Fig. 1 presents a schematic diagram of a rotary kiln–induction furnace process.

Coal acts as the source of both energy and reductant (carbon). Dolomite absorbs the sulfur and prevents contamination of sponge iron due to sulfur in the coal. The raw materials are heated to 1000 °C. The rotation and inclination of the kiln move the solids in the kiln towards the discharge end, while the pressure difference causes the motion of gases towards the feed end, creating a counter-current flow. Rotation of the kiln mixes the feed material and traps counter-current moving air to form reducing gas (carbon monoxide, CO) that reduces the iron ore (Fe_2O_3) to metallic iron. Through this counter-current arrangement, a solid mixture containing sponge iron, calcined dolomite, coal char, and ash emerges from the rotary kiln through the discharge end, and waste gases leave through the feed end. Subsequently, the solid mixture is cooled in the rotary cooler to 110 °C, followed by the separation of sponge iron using a magnetic separator. Cooling of sponge iron serves two purposes: (i) prevents reoxidation of sponge iron, and (ii) enables separation of sponge iron through a magnetic separator (iron has poor magnetic properties at high temperatures). Sponge iron from the magnetic separator is melted in an induction furnace to make steel. Dolo-char (mixture of calcined dolomite and coal char) can either be combusted in the captive power plant after blending with high volatile matter coal or sold to the market for energy uses or used as a landfill. Heat is recovered via the combustion of unburnt hydrocarbons, contained in the kiln off-gas, in a waste heat recovery boiler to generate electricity.

The major reactions taking place in the rotary kiln are described in Equations (1)–(4).

$$Fe_2O_3 + CO \rightarrow 2FeO + CO_2 \quad (1)$$

$$FeO + CO \rightarrow Fe + CO_2 \quad (2)$$

$$C + O_2 \rightarrow CO_2 \quad (3)$$

$$CO_2 + C \rightarrow 2CO \quad (4)$$

The mass balance of a rotary kiln-induction furnace process is represented through Equations (5) and (6) and the energy balance is represented with Equations (7) and (8). Equations (5) and (7) describe iron making in a rotary kiln, while equations (6) and (8) represent steel making in an induction furnace.

$$M_{ore} + M_{coal} + M_{dolomite} + M_{air} = M_{sponge\ iron} + M_{dolo-char} + M_{waste\ gas} + M_{dust} \quad (5)$$

$$M_{sponge\ iron} = M_{crude\ steel} + M_{slag} \quad (6)$$

$$Q = \sum m_2 h_2 - \sum m_1 h_1 + \Delta H_{reduction} + \Delta H_{calcination} + \Delta H_{loss} \quad (7)$$

$$Q = \sum m_2 h_2 - \sum m_1 h_1 + \Delta H_{loss} \quad (8)$$

In Equations (7) and (8), m_1 is the mass going into the reactor; m_2 is the mass coming out of the reactor; h_1 and h_2 are specific enthalpies associated with m_1 and m_2. To establish the mass balance, sponge iron required for producing one tonne of crude steel is computed based on the metallic content of sponge iron. Sponge iron quantity is used to calculate the ore required. The dolomite requirement is calculated based on the sulfur absorption capacity of dolomite and the quantity of coal used. The amount of coal used is computed based on the energy requirement from Equation (7).

All calculations are performed for a rotary kiln with a capacity of 350 tonnes per day. It is assumed that the ore is the only iron-containing feed material (using iron ore pellets is not considered). The rotary kiln is assumed to operate at 90 % of its rated capacity. The skin temperature (external surface temperature) of the rotary kiln is 300 °C. Solid material comes out of the kiln at 1054 °C and the off-gases leave the kiln at 851 °C. The iron ore fed to the rotary kiln achieves a metallization of 91 %, and the sponge iron contains 0.1 % carbon. It is also assumed that 60 % of the feed coal volatile matter does not combust inside the rotary kiln due to air deficiency. The energy and mass balances for a 350 t/d rotary kiln are presented in Figs. 2 and 3. The details of raw material composition, cost, and calculations are presented in the supplementary information file (see Tables S1–S8).

A typical sponge iron plant combusts dolo-char in the captive power plant. However, it is assumed that the power plant runs on waste heat recovery only and the dolo-char doesn't have energy use in the present study. While both feed coal and slinger coal release energy inside the rotary kiln, slinger coal is primarily responsible for maintaining the temperature profile inside the kiln. Due to the tumbling effect of the rotary kiln, fines are generated and carried away by dust leading to iron ore loss of 14.7 % and coal dust loss of 30 %. As the slinger coal has a lower residence time inside the rotary kiln than the feed coal, it is assumed that the char coming out as dolo-char is sourced from the slinger coal. All the carbon in char formed from the feed coal is consumed in reactions due to the comparatively higher residence time of feed coal. The char coming out as dolo-char has a carbon content of 22 %, and the dust coming out of the rotary kiln has a carbon content of 12 %. The fixed carbon content of coal is used for determining the proportion of feed coal and slinger coal. The ratio of feed coal to injection coal is 1.25. The mass and energy balances presented in this section help in establishing the CO_2 emissions across major steps of the overall process (Table 2). It may be noted that the methane emissions from the rotary kiln and other processes are not included in the present work due to the lack of data. The mass and energy balances presented in this section form the basis for calculating the emission reduction potential of various decarbonization measures discussed in the following section.

Table 1
Chemical composition of coal-based sponge iron lumps from the rotary kiln [42].

Grade	Grade 1	Grade 2	Grade 3
Total Iron (%, minimum)	89–93	88	85–88
Metallic iron (%, minimum)	80	76	70
Metallization (%, minimum)	88	86	82
Carbon content (%)	0.08–0.12	0.08–0.12	0.06–0.12
Sulfur content (%)	0.03	0.03	0.03
Phosphorus content (%)	0.06	0.07	0.10
$SiO_2 + Al_2O_3$ max (%)	6	7	8
Combined total of Pb, Zn, Cu, (%)	0.01	0.015	0.015

4. Decarbonization options

Iron-making is the most energy- and emission-intensive step across all processes in the iron and steel industry because iron-making requires energy for material handling, heating of raw materials, and reduction. In the rotary kiln process, the energy needed for reduction is provided by carbon, which makes the process more carbon-intensive. In the case of

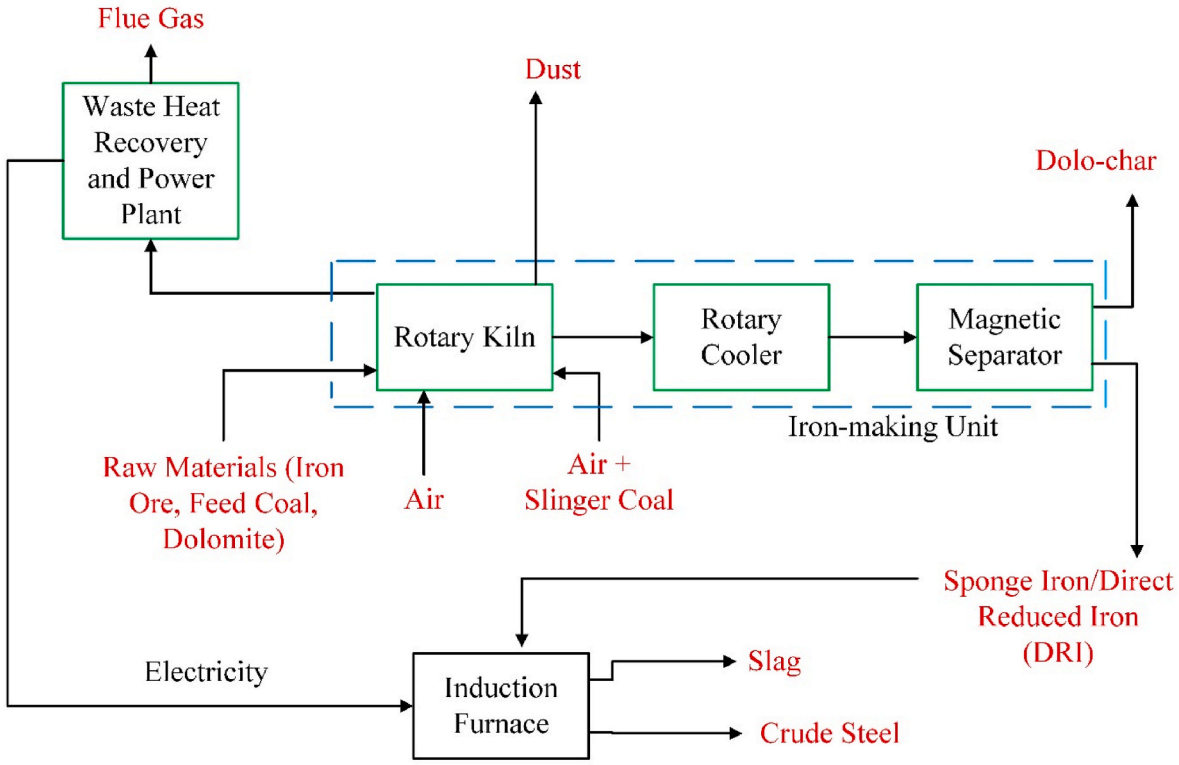


Fig. 1. Schematic diagram of a rotary kiln–induction furnace process for sponge iron and steel production.

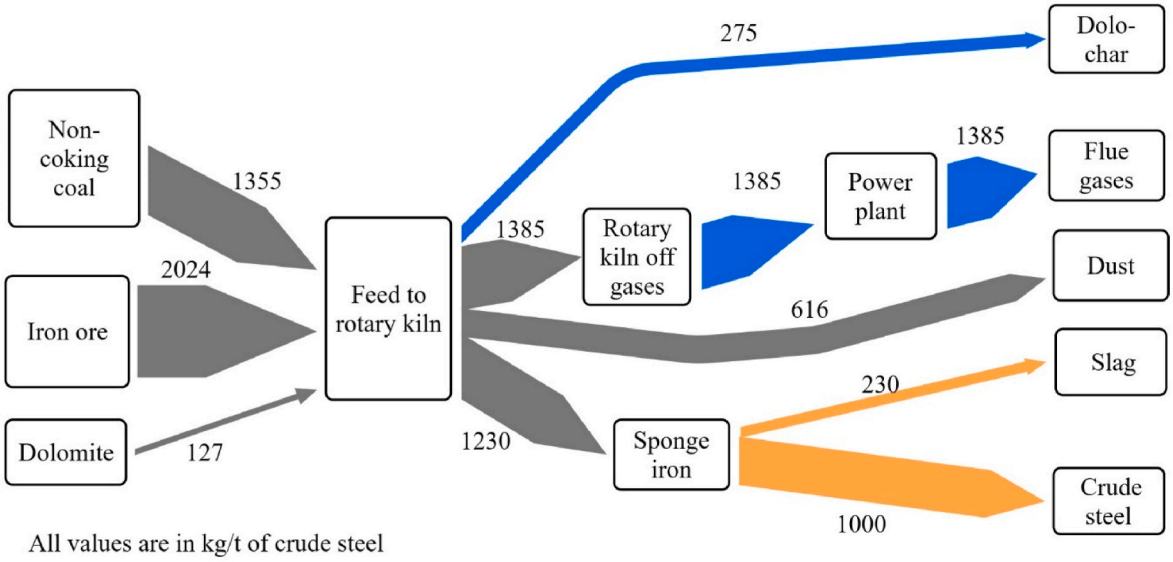


Fig. 2. Sankey diagram showing the mass balance of a rotary kiln–induction furnace process (basis: 1 tonne of crude steel).

rotary kiln–induction furnace route, steel-making primarily requires energy to melt sponge iron to enable conversion to steel. Various methods, including energy conservation, material replacement, fuel substitution, and renewable alternatives, can be adopted to decrease the emission intensity. Major CO_2 abatement measures are discussed below, and associated costs for each technology option are summarized in Table 3. It may be noted that the CO_2 abatement potential for each measure is estimated based on the mass and energy balances described in the previous section.

In Table 3, positive power saving means power output is increased in the process, and negative power saving means power output is decreased or power is consumed in the process. The operating cost is computed by adding the changes in electricity cost, cost of coal, and alternative material cost. Negative operation cost means savings are incurred during the operational phase of the process. The annualized fixed cost is calculated by multiplying the investment cost by the capital recovery factor. An annual discount rate of 10 % is considered in this work. Annual maintenance cost is assumed to be 10 % of the fixed cost. The total cost of an option is obtained by adding annualized fixed cost, annual maintenance cost, and operating cost.

Among the major decarbonization options discussed, painting of kiln, variable frequency drive, gravimetric separator, and char recycling

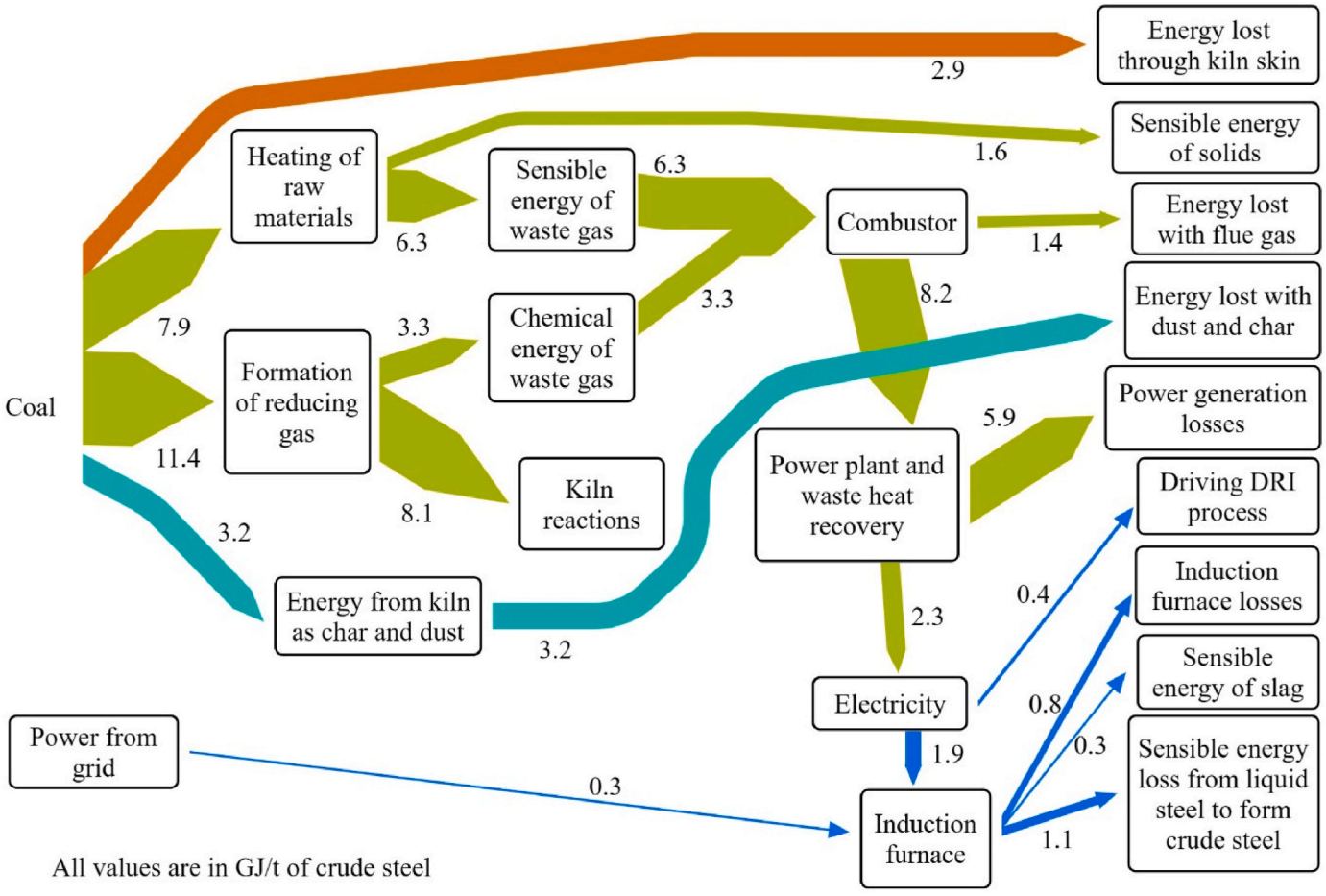


Fig. 3. Sankey diagram showing energy balance of a rotary kiln–induction furnace process (basis: 1 tonne of crude steel).

Table 2
CO_2 emissions from a rotary kiln–induction furnace process (basis: 1 tonne of crude steel).

Process steps	CO_2 emissions (kg CO_2/tcs)	CO_2 emissions (%)
Coal combustion (rotary kiln)	2114.1	97.4
Dolomite calcining	55.8	2.6
Total Process CO_2 emission	2169.9	100.0
Power usage (import)	67.2	
Effective CO_2 emissions	2237.1	

are energy efficiency options; the use of better grade dolomite is a material efficiency option; the use of biochar and bio-syngas are examples of renewable energy; and the injections of natural gas or hydrogen are examples of fuel substitution. Painting of kiln, variable frequency drive, and gravimetric separator have a negative cost of CO_2 abatement. This implies that the decarbonization through these options is economically beneficial. The rest of the options require a net investment.

It should be noted that the use of bio-syngas, natural gas, and green hydrogen are mutually exclusive options as only one of them can be implemented. Additionally, implementing any of these options eliminates the formation of char and thereby, eliminates the char recycle option. These four measures are represented as 6a, 6b, 6c, and 6d in Table 3 indicating that they are mutually exclusive.

4.1. Painting of kiln

The temperature inside the rotary kiln is high (900–1170 °C) [29], and results in a higher temperature of 300 °C [43] on the outer surface of the rotary kiln (also called kiln skin). High skin temperature results in more convective and radiative heat losses through the kiln surface. Surface heat loss can be reduced by painting the external surface of the rotary kiln with low-emissivity paint, which changes the skin temperature, as given in Equation (9). Equation (9) is obtained by equating the conduction heat transfer from the inner surface to the outer surface of the kiln wall with the sum of convection and radiation losses from the outer surface.

$$\sigma\varepsilon(T_2)^4+\left[h_c+\frac{2\left(\frac{K}{D}\right)}{\ln\left(\frac{D}{2R_1}\right)}\right]T_2-\left[\sigma\varepsilon(T_o)^4+h_cT_o+\frac{2\left(\frac{K}{D}\right)}{\ln\left(\frac{D}{2R_1}\right)}T_1\right]=0 \quad (9)$$

In Equation (9), σ is the Stefan–Boltzmann constant, ε is the emissivity of the kiln surface, T_2 is the skin temperature (K), h_c is the convective heat transfer coefficient, K is the thermal conductivity of the rotary kiln wall, D is the external diameter of the rotary kiln, R_1 is the internal radius of the rotary kiln, T_0 is the ambient temperature (K), and T_1 is the temperature inside the rotary kiln (K).

The amount of heat loss before and after painting the rotary kiln is estimated via Equation (10) by changing the emissivity for various cases and the surface temperature obtained for a particular emissivity from Equation (9).

$$\dot{Q}=\sigma\varepsilon\pi DL\left[(T_2)^4-(T_0)^4\right]+h_c\pi DL(T_2-T_0) \quad (10)$$

L is the length of the rotary kiln, and $\dot{Q}$ is the rate of heat loss through the kiln skin in Equation (10).

The energy saved is calculated based on the difference in heat loss

Table 3
Major carbon abatement options and associated impact on rotary kiln–induction furnace process of crude steel production.

#	Abatement options (technology status)	Direct CO_2 abatement (kg/tcs)	Power savings (kWh/tcs)	Net CO_2 abatement (kg/tcs)	Fixed cost ($/tcs)	Maintenance cost ($/tcs)	Operation cost ($/tcs)	Total cost ($/tcs)	Cost of abated carbon ($/t CO_2)
1	Painting of kiln (proven)	3.9	0.0	3.9	0.0	0.0	−0.1	−0.1	−33.3
2	Variable frequency drive (proven)	0.0	29.9	24.5	0.2	0.0	−2.5	−2.3	−93.8
3	Gravimetric separator (to be tested)	0.0	175.9	67.2	0.0	0.0	−11.4	−11.4	−169.3
4	Better grade dolomite (proven)	56.6	0.0	56.6	0.0	0.0	2.9	2.9	51.7
5	Biochar for feed coal (tested on pilot scale)	1241.5	378.7	1308.7	17.0	1.7	72.2	90.9	69.4
6a	Char recycle (proven)	26.5	−35.2	−2.4	0.0	0.0	−1.4	−1.4	567.3
6b	Bio-syngas for slinger coal (tested on pilot scale)	872.6	−66.1	818.4	2.2	0.2	28.5	30.9	37.8
6c	Natural gas injection (proven)	442.5	−66.1	388.3	0.0	0.0	60.0	60.0	154.4
6d	Green hydrogen injection (to be tested)	872.6	−66.1	818.4	0.0	0.0	320.8	320.8	392.0

from kiln skin and injection coal saved. The coal combustion efficiency and the calorific value of coal are used to estimate the quantity of coal savings. The carbon abatement cost for this option is −33.3 $/t CO_2, where the negative cost signifies a net saving associated with the implementation of this option.

4.2. Variable frequency drive

The rotary kiln uses both iron ore lumps and iron ore pellets as feed material. The typical capacity utilization of the sponge iron-making unit is 90 % with iron ore lumps. Variable frequency drives can help reduce electricity consumption as equipment running at 90 % of the rated speed consumes only 73 % of the rated power [44]. Motors power the conveyor system, rotation of the rotary kiln and rotary cooler, coal cutter, screens, compressors, and pumps. It is assumed that 90 % of the electricity is consumed by motors. The amount of CO_2 saved is computed by multiplying the electricity savings with the grid emission intensity (see Table 3). The negative carbon abatement cost of −93.7 $/t CO_2 signifies a net saving associated with the implementation of this option.

4.3. Gravimetric separator

In a traditional rotary kiln process, sponge iron comes out of the rotary kiln at about 1020 °C and is then cooled to 110 °C in a rotary cooler before being separated from a solid mixture containing sponge iron, coal ash, and dolo-char mixture. Sponge iron is then stored and later used in the steel-making process. The sponge iron is reactive and the heat recovery from the rotary cooler is avoided in a conventional plant to prevent its reoxidation. The rotary cooler can be eliminated, and the magnetic separator can be replaced with a gravimetric separator to separate the sponge iron at high temperatures. Hot sponge iron can be directly charged into the induction (electric) furnace, resulting in electricity savings. The amount of electricity saved in the induction furnace is calculated based on the heat content of sponge iron at the time of charging it into the induction furnace. It is assumed that the sponge iron retains 85 % of its sensible energy at the time of charging in the induction furnace. The net electricity savings are calculated by adding electricity savings in the induction furnace unit, electricity savings due to the elimination of cooler, water pump, and magnetic separator and subtracting the electricity consumed by the magnetic separator.

Table 2 suggests that only 67.2 kg of CO_2 is attributed to power import (from the grid) which is due to 82 kWh of imported power with a grid emission intensity of 0.82 kg CO_2/kWh. The net power savings by implementing gravimetric separator is 175.9 kWh/t of crude steel which is significantly higher than the power bought from the grid. The CO_2 savings is calculated considering only the avoidance of power purchased from the grid. Surplus power sold to the grid generates revenue but doesn't decrease CO_2 emissions. Note that the grid supplies electricity at a higher cost of 8.4 ¢/kWh and purchases electricity at a lower cost of 4.8 ¢/kWh, as listed in Tables S5 and S6 of the supplementary information file. The operating cost represents the net effect of avoiding electricity purchase and selling surplus electricity to the grid. A negative carbon abatement cost of −169.3 $/t CO_2 is calculated for this option.

4.4. Better-grade dolomite

Dolomite helps in absorbing coal sulfur in the rotary kiln. Otherwise, coal sulfur is absorbed by sponge iron and results in contamination. Note that dolomite gets calcined due to the high temperature inside the rotary kiln and releases CO_2. Calcined dolomite contains calcium oxide (CaO), which absorbs sulfur. However, the share of calcium oxide in limestone/dolomite varies from 20 % to 60 %, depending on its source of origin [45]. The use of better-grade dolomite in rotary kilns is helpful in decreasing the CO_2 emissions due to calcination. A positive carbon abatement cost of 51.7 $/t CO_2 signifies a net investment associated with the implementation of this option.

4.5. Biochar replacing feed coal

High temperature inside the rotary kiln results in the conversion of feed coal to coal char. Due to the reactivity of char and deficient air supply, carbon monoxide (CO) is formed which helps in the iron ore reduction process. Feed coal can be replaced by biochar obtained from woody biomass. Heating woody biomass in the absence of air produces biochar and other combustible products. Due to the higher reactivity of biochar to form carbon monoxide as compared to coal char [46], biochar can fully replace feed coal in the rotary kiln process. Feed coal is substituted by biochar on an energy-equivalent basis. Biochar from renewable plantations eliminates the use of fossil carbon in sponge iron and crude steel production. The use of biochar in rotary kilns decreases the quantity of unburnt hydrocarbons in the after-burning chamber, which further leads to a decrease in the power output from the power plant. However, combustible byproducts generated during the charring of woody biomass are combusted to generate power resulting in a net increase in power by 378.7 kWh/t of crude steel. The use of biochar in a rotary kiln does not pose any significant technological challenges [47]. It was also tested on a pilot scale by one of the Indian steel producers. However, this option faced challenges as the supply chain for biomass is not yet well developed due to its distributed and seasonal availability.

The amount of CO_2 savings is calculated by adding the emissions

eliminated due to feed coal and grid electricity. Surplus power is exported to the grid and is accounted for the revenue generation, but not in CO_2 savings as is the case with gravimetric separator. This option has the highest CO_2 abatement potential and can eliminate 1308.7 kg CO_2/tcs.

4.6. Char recycling

The magnetic separator separates sponge iron and dolo-char (a mixture of calcined dolomite and char). Dolo-char can be separated using a gravimetric separator, and the coal char can be recycled as a feed material replacing an equivalent amount of feed coal. The amount of feed coal replaced is calculated by equating the fixed carbon in coal char with fixed carbon in coal. Reduction in feed coal decreases the CO_2 emissions, however, it also decreases the power output from the waste heat recovery and power plant. The power output is decreased because feed coal releases volatile matter, most of which goes to the waste heat recovery boiler uncombusted. It is assumed that only 90 % of the char is separated from the dolo-char mixture. The amount of CO_2 saved is calculated by adding the emissions avoided due to decreased coal usage, the amount of CO_2 released from char combustion, and the CO_2 emission equivalent due to power loss in the power plant. The power required to separate char from the dolo-char mixture is also accounted for as power loss. The operation cost is computed by subtracting the cost of total power lost from the cost of coal saved due to the implementation of char recycling. Char recycling is not a CO_2 abatement option when implemented independently (see Table 3) because the direct carbon abatement is just 26.5 kg CO_2 with a power loss of 35.2 kWh/tcs. This power loss is compensated by grid electricity and results in a net increase in CO_2 emissions by 2.4 kg/t of crude steel. However, in the case of combined implementation of several decarbonization options (as discussed later), power purchase from the grid is eliminated, and direct CO_2 abatement becomes equal to total CO_2 abatement.

4.7. Bio-syngas replacing slinger coal

Slinger (injection) coal maintains the temperature inside the rotary kiln. It can, therefore, be replaced with gaseous fuel for higher combustion efficiency and decreasing the net energy input. Biomass can be gasified in a gasifier, and the bio-syngas can be injected through the discharge end instead of slinger coal. This option eliminates the use of slinger coal and also decreases the power output from the power plant because higher combustion efficiency of gaseous fuel decreases the quantity of uncombusted kiln off-gas going into the power plant. The use of coal syngas has been successfully tested by small sponge iron plants. However, it is not practiced at present due to significant differences in the run time of the gasifier compared to the campaign life of a rotary kiln. Renewable biomass is a carbon-neutral option, and hence, the use of bio-syngas to replace coal is considered a decarbonization option.

4.8. Natural gas replacing slinger coal

Coal is a solid fuel and has poor combustion efficiency as compared to natural gas. Further, natural gas has lower CO_2 emissions per unit of energy than coal. This makes natural gas an excellent alternative to slinger coal. Similar to the bio-syngas option, it also decreases power output from the power plant due to the decreased quantity of uncombusted/partially combusted components in the kiln off-gas. Typical continuous run-time (also called campaign life) of a rotary kiln is usually 110 days as compared to the campaign life of gasifiers which is in the range of 20 days. Due to this, the implementation of gasifier-based carbon abatement options poses operational issues. Natural gas injection to replace slinger coal does not have any such operational problems and can be implemented easily and reliably. However, natural gas provides lower CO_2 abatement as compared to carbon-neutral bio-syngas.

4.9. Green hydrogen replacing slinger coal

A rotary kiln primarily uses coal as a reductant. Coal is fed into the rotary kiln through two different streams, feed coal and slinger coal. Feed coal goes into the kiln along with iron ore and dolomite through the feed end. Due to the rotary motion of the kiln (~0.5 rpm) and inclination with the horizontal (2.5–3 %), the feed materials get mixed together and move towards the discharge end. The rotary kiln has two phases: oxidizing gas phase and reducing bed phase. In the gas phase, volatiles and carbon monoxide react to form carbon dioxide. Due to the rotary motion of the kiln, CO_2 is trapped in the bed phase, where it reacts with solid carbon to form carbon monoxide, and reduces Fe_2O_3 to metallic iron. Owing to this process requirement, feed coal is not replaced. On the other hand, slinger coal is fed to the kiln through the discharge end to maintain a high temperature inside the rotary kiln. In this study, the slinger coal is replaced with green hydrogen for decarbonization.

A methodology for calculating the CO_2 abatement is discussed in the following section considering the interdependency of the various decarbonization options.

5. Abatement and green premium curves

Any two decarbonization measures can either be independent (i.e., they do not impact each other), dependent (i.e., they may impact each other but they can be implemented simultaneously), or mutually exclusive (i.e., they cannot be implemented simultaneously). This interdependency plays a major role in determining the abatement cost curve for decarbonization. Krishna Priya et al. [48] described an algorithm to arrange such options. Another methodology to determine the abatement cost curve with such constrained options is discussed as follows.

5.1. Methodology

A flowchart, highlighting the sorting of decarbonization options, is presented in Fig. 4. To visualize the effect of all options, the complete algorithm needs to be reiterated with each set of mutually exclusive options such that none of the options in a particular set are mutually exclusive. In the case of dependent options, only the marginal effect is assigned to the option implemented later. Marginal values of cost and emission are calculated by subtracting the impact of the better option (with lower CO_2 abatement cost) from the combined implementation of both options.

At the beginning of the sorting algorithm, two sets are assumed. The first set (set A) has all the decarbonization options placed in it and the second set (set B) is empty. At the beginning of the first cycle, all the options in set A are sorted in increasing order of the cost of carbon abatement. The best option is moved to set B. The cost and CO_2 abatement of each option in set A is reevaluated considering the selection of options transferred to set B. If any option in set A is mutually exclusive to options in set B, that option is discarded. If an option in set A is independent of options in set B, then the cost and CO_2 abatement for that option does not change. Subsequently, the options in set A are again sorted to identify the option with the least carbon abatement cost and the cycle is repeated till the last option in set A is moved to set B. During each cycle of repeating the algorithm, one option is moved from set A to set B. The algorithm ends when the set A becomes empty.

All the sorted options are used to plot the carbon abatement cost curve. If an option does not have any effective carbon abatement but decreases the cost, it is removed while generating the marginal abatement cost curve. However, it may still be a part of the green premium curve as discussed later.

5.2. Marginal carbon abatement curve

Carbon abatement options, listed in Table 3, are sorted (as proposed

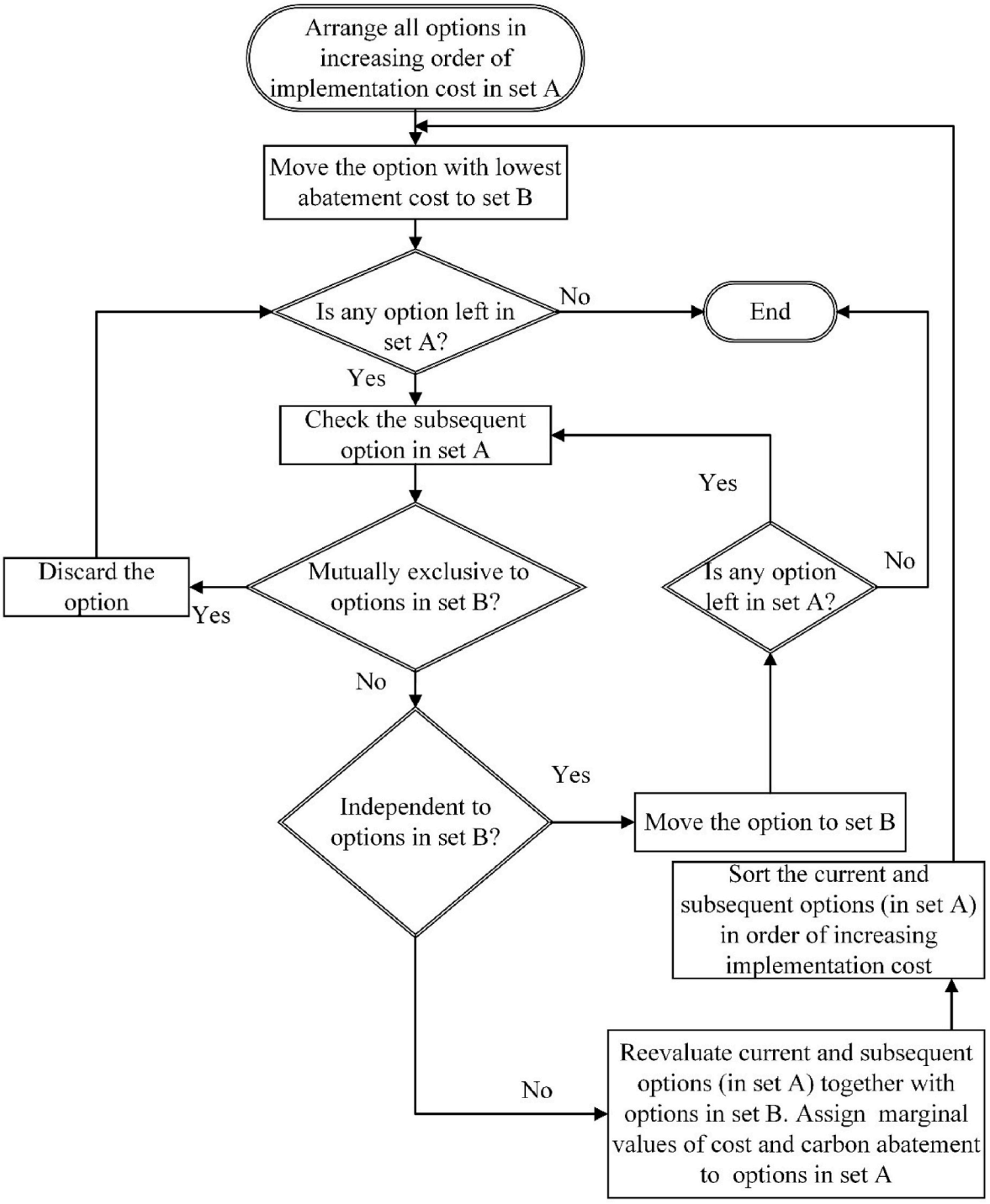


Fig. 4. Flowchart to sort decarbonization options in order of implementation priority.

in Fig. 4) to generate the marginal carbon abatement curve. The option with the lowest abatement cost is selected first. Among the options listed in Table 3, the gravimetric separator (option 3) is identified to be the cheapest and selected as the first option for decarbonization. Then the impact of implementation of gravimetric separator on cost and carbon abatement of subsequent options are reevaluated (as listed in Table S9 of the supplementary information file).

The power savings due to the gravimetric separator eliminate power purchase from the grid and therefore, no carbon credit can be given to the electricity savings from the variable frequency drive. As there is no carbon abatement, the variable frequency drive option (option 2) does not have any effect on the marginal abatement cost curve. However, it is still selected as the net cost is negative (see Table S9) and it provides an economic advantage, especially for determining the green premium curve (discussed in the following sub-section). The selection of the option of variable frequency drive does not have any impact on subsequent options as it is independent of other options, and therefore, the order of the other options remains unaltered.

The selection of a gravimetric separator annihilates the requirement of a char separator for char recycling, reducing the fixed cost for the char recycling option (option 6a) to zero. Furthermore, the emission penalty due to the additional power requirement for this option decreases to

make this option a net carbon abatement option (Table S9). The char recycling is selected as the third option. The bio-syngas replacing slinger coal (option 6b), natural gas injection (option 6c), and hydrogen injection (option 6d) options are mutually exclusive to the char recycling (option 6a) option and are, therefore, discarded.

Table S10 in the supplementary information file lists the updated cost and emission abatement values for the remaining options after selecting the gravimetric separator, variable frequency drive, and char recycling. The next best option is the painting of the kiln and it is selected as the fourth option. This option (i.e., the painting of the kiln) is independent of the subsequent options, and therefore, the order of the better-grade dolomite and the biochar replacing feed coal remains unchanged (Table S10). The use of better grade dolomite is the next best option and it does not impact the biochar option. Finally, biochar replacing feed coal is selected as the last option. These six options constitute the marginal abatement cost curve, as shown in Fig. 5.

It is evident from Fig. 5 that gravimetric separator, char recycle, and painting of kiln have negative carbon abatement costs. This is because these are energy efficiency options - gravimetric separator decreases the electricity requirement in the induction furnace, char recycle and painting of the kiln decrease the coal input to the rotary kiln. By implementing these six options, 1247.6 kg CO_2/t of crude steel can be abated in total and 55.8 % of CO_2 emission can be abated. It may be noted that 97.6 kg CO_2/t of crude steel can be abated with a net cost benefit using the three energy efficiency options. Biochar replacing feed coal provides the maximum CO_2 abatement of 1241.5 kg CO_2/t of crude steel. This is due to the carbon neutral consideration of biochar. Variable frequency drive doesn't have an effective CO_2 abatement and therefore it is not represented in the curve (Fig. 5).

The complete algorithm is repeated without considering the char recycle option (option 6a in Table 3). As the char recycling (option 6a), bio-syngas replacing slinger coal (option 6b), natural gas injection (option 6c), and green hydrogen injection (option 6d) options are mutually exclusive, four different abatement curves are generated by excluding one after another. The updated cost and carbon abatement values for the selections containing bio-syngas replacing slinger coal, natural gas injection, and green hydrogen injection are presented in Tables S11, S12, and S13 in the supplementary information file. The overall CO_2 abatement depends on the selection of decarbonization options. The marginal abatement cost curves considering biochar replacing feed coal, natural gas injection, and green hydrogen injection are shown in Fig. 6.

For the three decarbonization pathways described in Fig. 6, only 71.1 kg of CO_2 can be abated with a net cost benefit by implementing a gravimetric separator and painting of kiln. However, the total CO_2 abatement possible with both these pathways is much higher: 2218.4 kg CO_2/t of crude steel for the bio-syngas route (99.2 % of CO_2 abatement potential), 1799.2 kg CO_2/t of crude steel for the natural gas route (80.4 % of CO_2 abatement potential), and 2218.4 kg CO_2/t of crude steel for the green hydrogen route (99.2 % of CO_2 abatement potential) due to the replacement of slinger coal with lower emission intensity fuels. Furthermore, natural gas, green hydrogen, and bio-syngas have better combustion efficiency of 95.0 % as compared to 70.8 % for slinger coal. Natural gas has lower CO_2 emission per unit of energy than that of coal, and green hydrogen and bio-syngas are considered carbon neutral – resulting in higher CO_2 abatement possibilities as compared to natural gas. Natural gas injection is considered despite having lower CO_2 abatement and higher cost than the bio-syngas option due to ease of implementation. Bio-syngas option requires a gasifier that has operational issues while natural gas can be used directly. For these three decarbonization pathways, biochar replacing feed coal provides the maximum carbon abatement followed by options of substituting slinger coal due to carbon-neutral consideration for biochar and green hydrogen.

5.3. Green premium curve

Green premium represents the change in the cost of crude steel due to

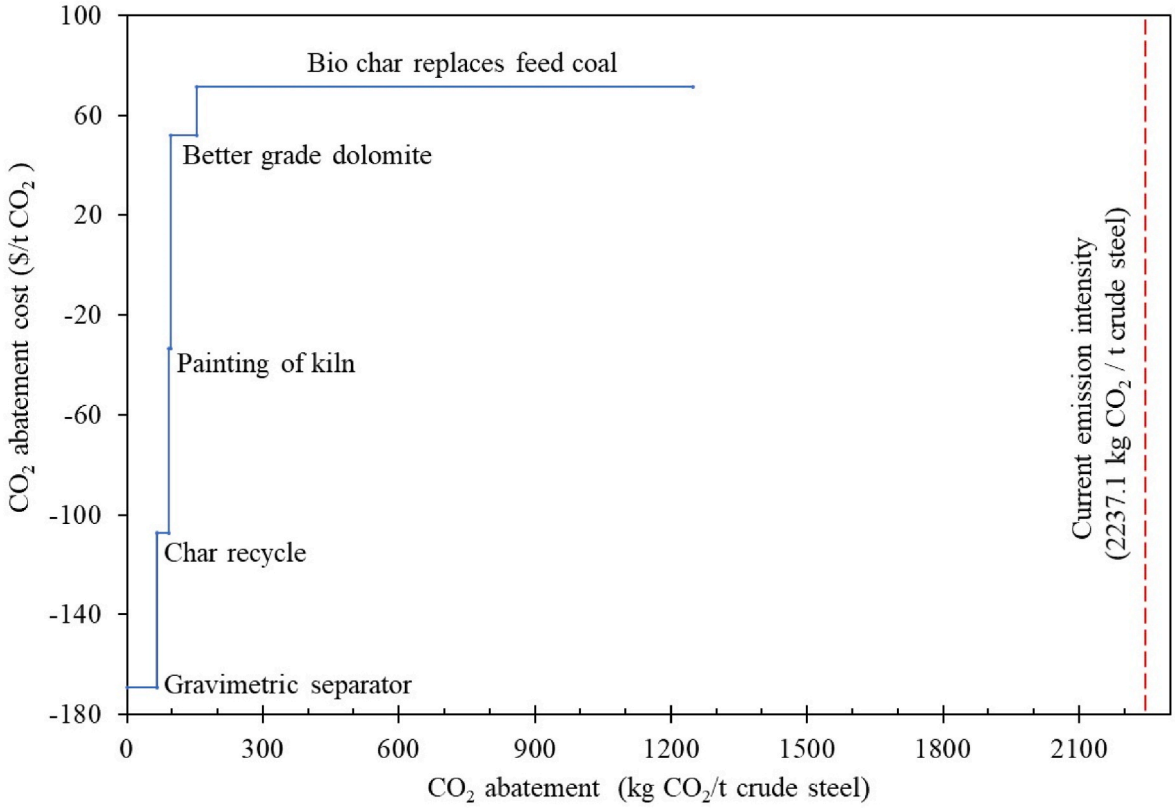


Fig. 5. Marginal abatement cost curve for various decarbonization measures (char recycle option pathway) for the rotary kiln induction furnace process.

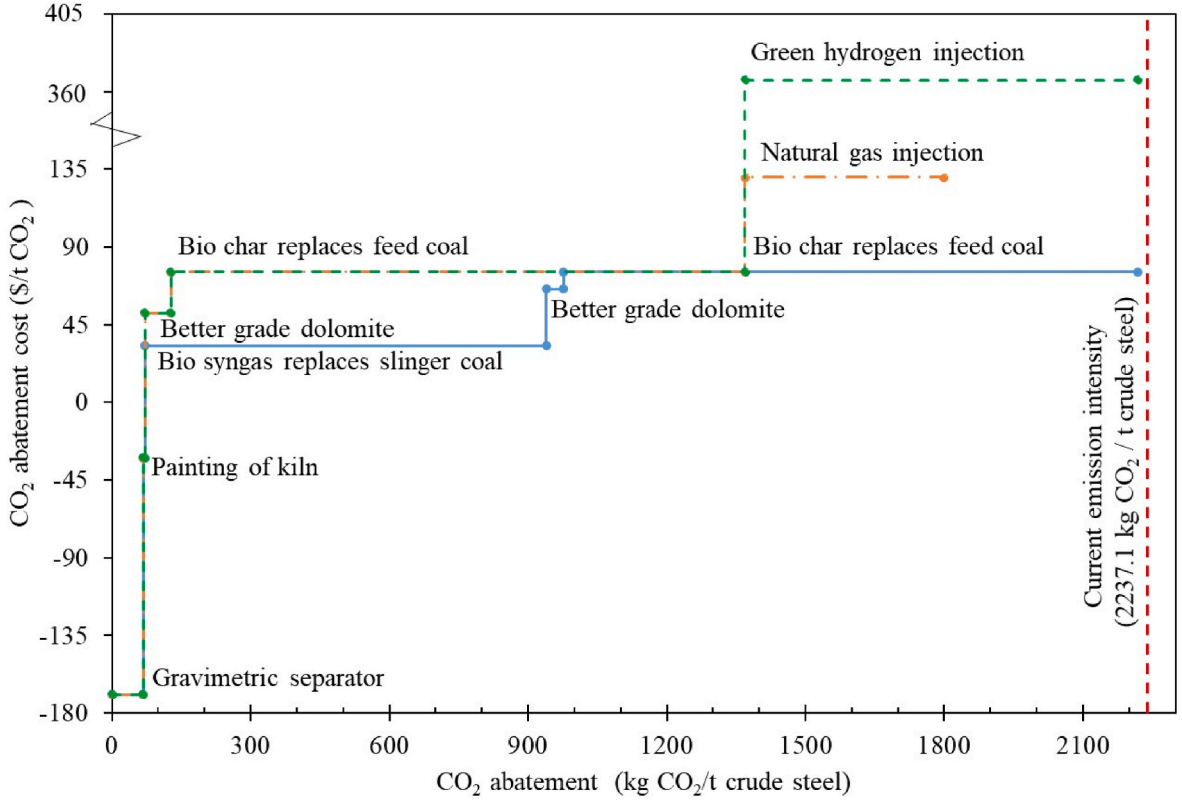


Fig. 6. Marginal abatement cost curves for various decarbonization measures (bio-syngas replacing slinger coal, natural gas injection, and green hydrogen injection pathways) for the rotary kiln induction furnace process.

the implemention of a carbon abatement option to the conventional process. If the carbon abatement cost is negative, the green premium cost is also negative. This implies that the cost of crude steel is expected to be reduced while emission reduction is achieved. For example, the implementation of a gravimetric separator (see Table 3) is expected to reduce the cost of steel by 11.4 $ and the CO_2 emission intensity by 67.2

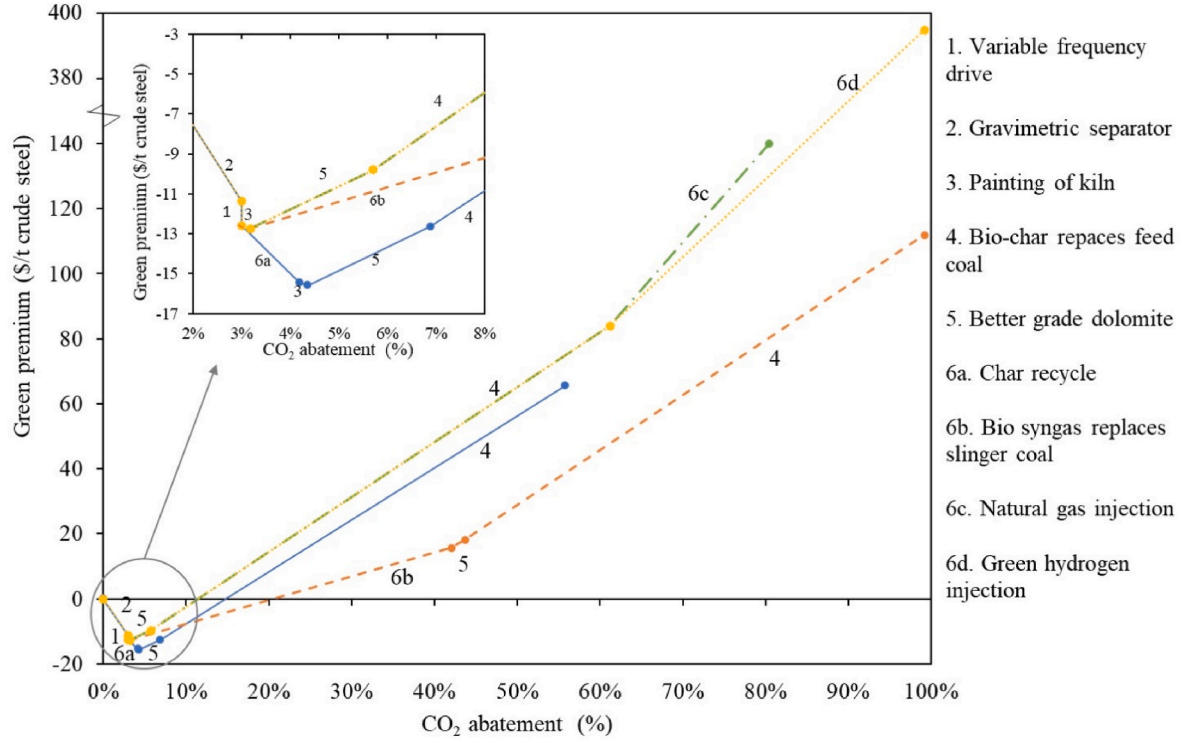


Fig. 7. Green premium curves for various decarbonization pathways for the rotary kiln induction furnace process.

kg on a per tonne basis. On the other hand, a positive carbon abatement cost implies an increase in the cost of crude steel required for emission reduction. For example, the use of better grade dolomite (see Table 3 and S10) is expected to increase the cost of steel by 2.9 $ and reduce the CO_2 emission intensity by 56.6 kg on a per tonne basis. As the options are selected for implementation in the order of increasing cost of carbon abatement, the green premium changes accordingly. For multiple carbon abatement options, a green premium curve represents the cumulative change in the cost of crude steel. Every marginal abatement cost curve translates into a green premium curve. Note that a marginal abatement cost curve represents the cost-effectiveness of various carbon abatement options and helps in selecting options that are economical for carbon abatement. On the other hand, a green premium curve represents the cumulative cost-effectiveness of these options and helps in determining the cumulative economic effect of these measures on crude steel prices.

The green premium curves for the rotary kiln induction furnace process, corresponding to the different marginal abatement cost curves, are shown in Fig. 7. It may be noted that even if an option does not have effective carbon abatement (for example, variable frequency drives), it can still be included in the green premium curve provided there are cost savings. The intersection of a green premium curve with the *x*-axis gives the amount of CO_2 that can be abated without a net cost to the customer.

Considering the implementation of a decarbonization pathway containing char recycling (Fig. 5), 14.8 % of the emissions can be abated without a net cost (see Fig. 7). This option can abate 55.8 % of emissions with a green premium of 65.5 $/t of crude steel. Selecting the decarbonization pathway with natural gas injection (Fig. 6), 11.5 % of the emissions can be abated without a net cost (see Fig. 7). 80.4 % of the emission can be abated with the natural gas injection pathway with a net green premium of 140.0 $/t of crude steel. Similarly, with the decarbonization pathway containing the bio-syngas option (Fig. 6), 20.6 % of the emissions can be abated without a net cost (see Fig. 7). This option can abate emissions by 99.2 % in total with a net green premium of 11.9 $/t of crude steel. The CO_2 emission due to the calcination of dolomite cannot be completely abated in this decarbonization pathway. Similar to natural gas injection, the selection of a decarbonization pathway with green hydrogen injection (Fig. 6) can help abate 11.5 % of the emissions without a net cost (see Fig. 7). However, this option can eliminate 99.2 % of the emissions at a green premium of 366.6 $/t of crude steel.

Implementation of the variable frequency drives does not have a direct impact on carbon abatement. However, due to the cost benefits, it increases the net zero green premium potential. A vertical drop in the green premium curve can be observed due to the adoption of this measure (see inset in Fig. 7). 1.7 % of additional emission abatement can be achieved, without additional cost to the customer, for the bio-syngas pathway due to the incorporation of the variable frequency drive option.

It is observed from Fig. 7 that the pathway containing the char recycling option is the most economical way for up to 9.9 % of CO_2 abatement, after which the bio-syngas pathway becomes the most economical. Considering gravimetric separator, variable frequency drive, painting of kiln, bio-syngas to replace slinger coal, and better grade dolomite, 43.7 % of CO_2 emissions (977.0 kg CO_2/t of crude steel) can be abated at a cost of 18.1 $/t of crude steel.

6. Conclusions

India is the largest producer of sponge iron globally – however, its production is associated with high energy and emission intensities due to the use of a coal-based rotary kiln process. The findings of this paper highlight the feasibility and effectiveness of implementing various CO_2 abatement options to reduce emissions in the rotary kiln induction furnace process. The results indicate that the choice of decarbonization pathways can impact emission reduction. Near-complete decarbonization of the coal-based sponge iron production and subsequent crude steel manufacturing has a very high green premium in the range of 112–395 $/t of crude steel for nearly 80 %–99 % of CO_2 abatement. However, a significant share of the emissions (11.5 %–20.6 % of the total decarbonization potential) can be abated without a net cost to the customer (i.e., net zero green premium) which underscores the economic viability of implementing certain strategies, like the use of variable frequency drives, kiln surface painting, use of better grade raw material (dolomite) and use of better equipment like a gravimetric separator for product separation.

The proposed decarbonization measures include improved energy efficiency, improved material efficiency, use of renewable energy, and fuel substitution. Through a combination of these measures, a substantial decarbonization of sponge iron industries can be achieved. By exploring the decarbonization potential, associated costs, and the synergistic effects of implementing multiple abatement options, this research contributes to the development of informed policy decisions. The insights gained from this research provide a foundation for designing and implementing a net-zero pathway for the sector. By incorporating the identified decarbonization strategies, policymakers can effectively mitigate the environmental impact of the sponge iron industry while promoting sustainable growth and resource efficiency. Industrial symbiosis can play a significant role in the decarbonization of the entire sector directly (e.g., dolo-char can also be used in the cement industry and can be helpful in decreasing otherwise emissions from calcination in clinker units) or indirectly (e.g., purchase of carbon credits). Future studies can delve deeper into specific technologies, explore potential barriers and enablers for implementation, and assess the long-term economic and environmental impacts of adopting various decarbonization options, including industrial symbiosis.

CRediT authorship contribution statement

Nishant Bhardwaj: Writing – original draft, Validation, Formal analysis. **Srinivas Seethamraju:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Santanu Bandyopadhyay:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Nomenclature

D – external diameter of the rotary kiln

DRI – direct reduced iron
h_1 – enthalpy of m_1
h_2 – enthalpy of m_2
h_c – convective heat transfer coefficient
K – Overall thermal conductivity of the rotary kiln wall
L – length of the rotary kiln
m_1 – mass going into the reactor
m_2 – mass coming out of the reactor
M_{air} – mass of air
M_{coal} – mass of coal
$M_{crude\ steel}$ – mass of crude steel
$M_{dolo\ -\ char}$ – mass of dolo – char
$M_{dolomite}$ – mass of dolomite
M_{dust} – mass of dust from kiln containing coal ash and unreduced iron ore fines
M_{ore} – mass of iron ore
M_{slag} – mass of slag
$M_{sponge\ iron}$ – mass of sponge iron
$M_{waste\ gas}$ – mass of waste gas
$\dot{Q}$ – heat loss through the kiln skin per second
R_1 – internal radius of the rotary kiln
tcs – tonne of crude steel
T_0 – ambient temperature, K
T_1 – temperature inside the rotary kiln, K
T_2 – skin temperature, K
ε – emissivity constant of the kiln surface
σ – Stefan–Boltzmann constant
$\Delta H_{calcination}$ – Energy required for calcination
ΔH_{loss} – Heat loss from the reactor
$\Delta H_{reduction}$ – Energy required for reduction

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2024.132516.

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